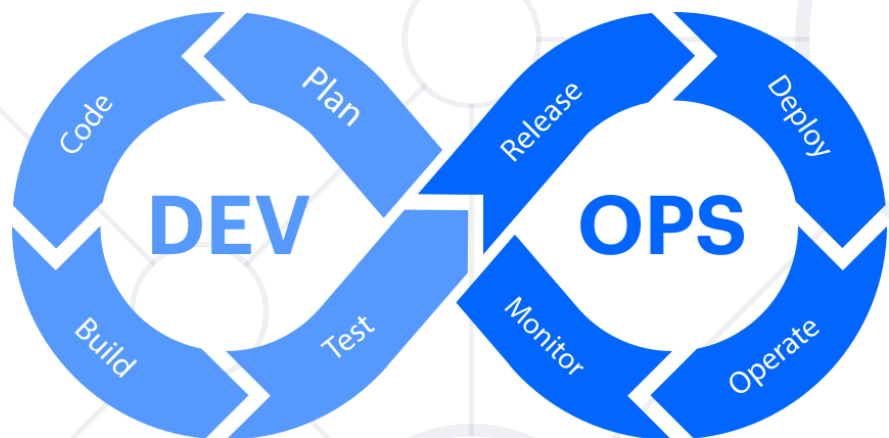


Testing

Strategy. Types. Automation



SoftUni Team
Technical Trainers



SoftUni



Software University

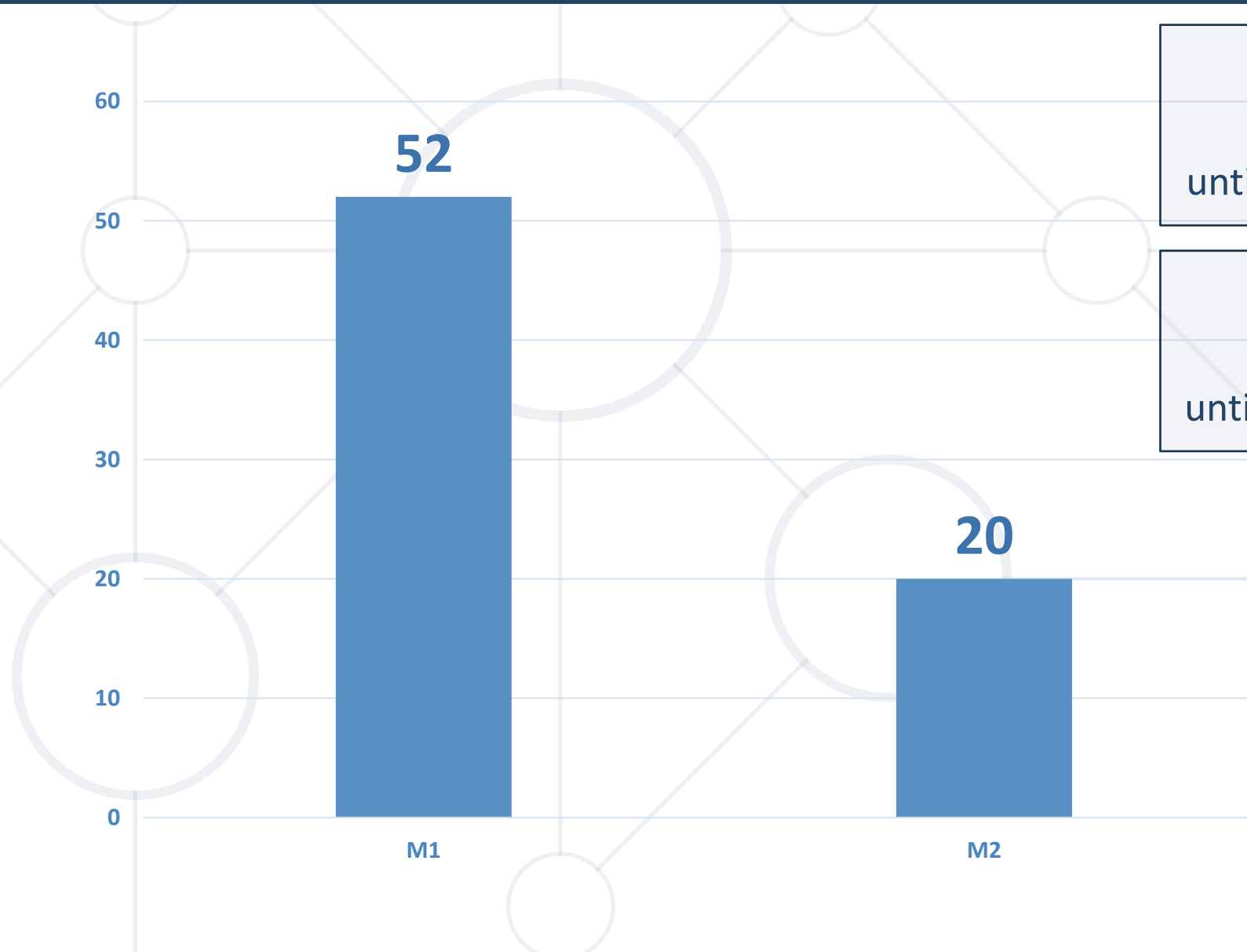
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#DevOps-CI-CD

**[facebook.com/groups/
cicdmonitoringjanuary2026](https://facebook.com/groups/cicdmonitoringjanuary2026)**

Homework Progress



Solutions for M2
can be submitted
until 23:59 on 04.02.2026

Solutions for M3
can be submitted
until 23:59 on 11.02.2026



Previous Module (M2)

Quick Overview

What We Covered

1. Version Control and Strategy
2. Workflow Selection
3. Build Triggers and Tools
4. Artifacts Management

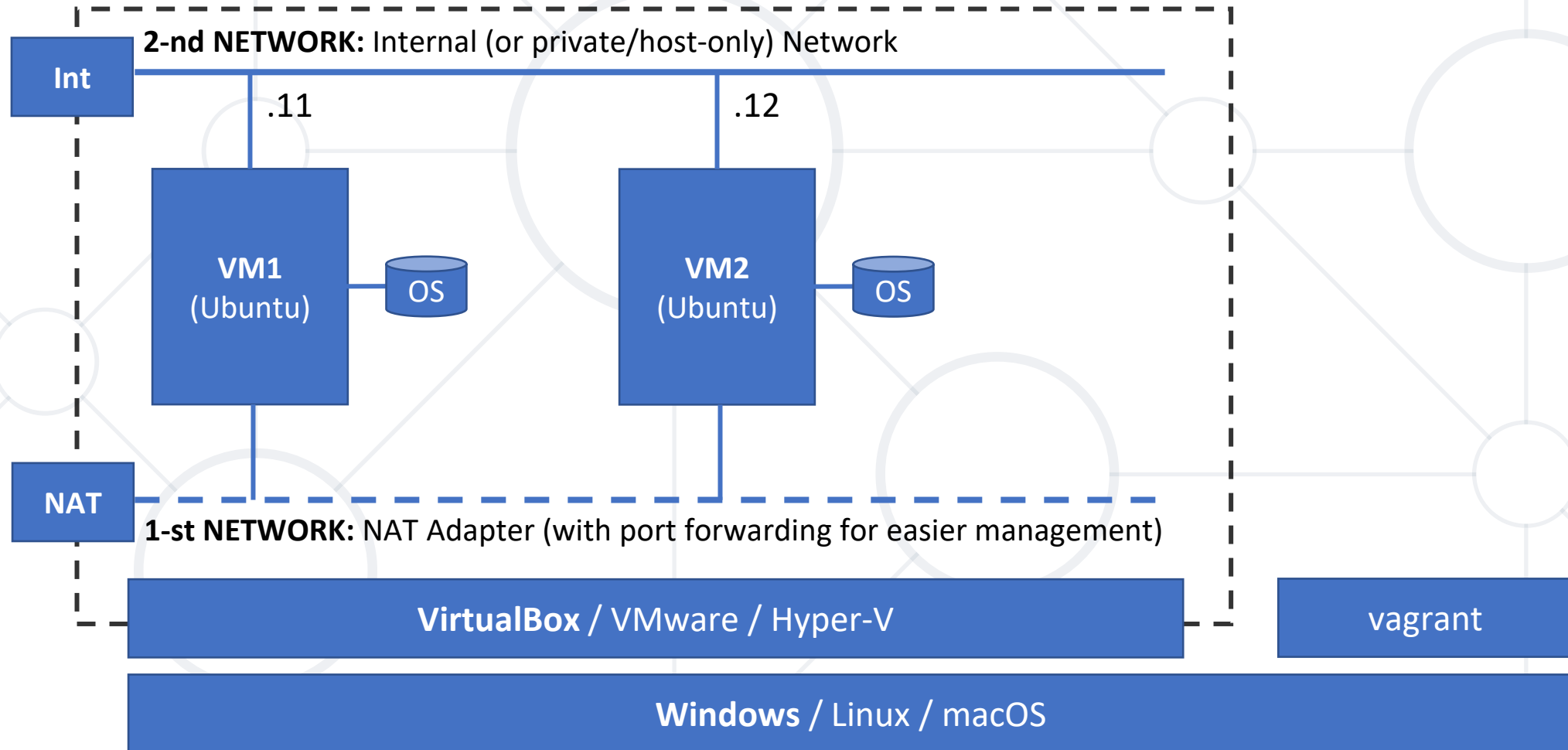


This Module (M3)
Topics and Infrastructure

Table of Contents

1. Testing Strategy
2. Test Types
3. Automated Testing





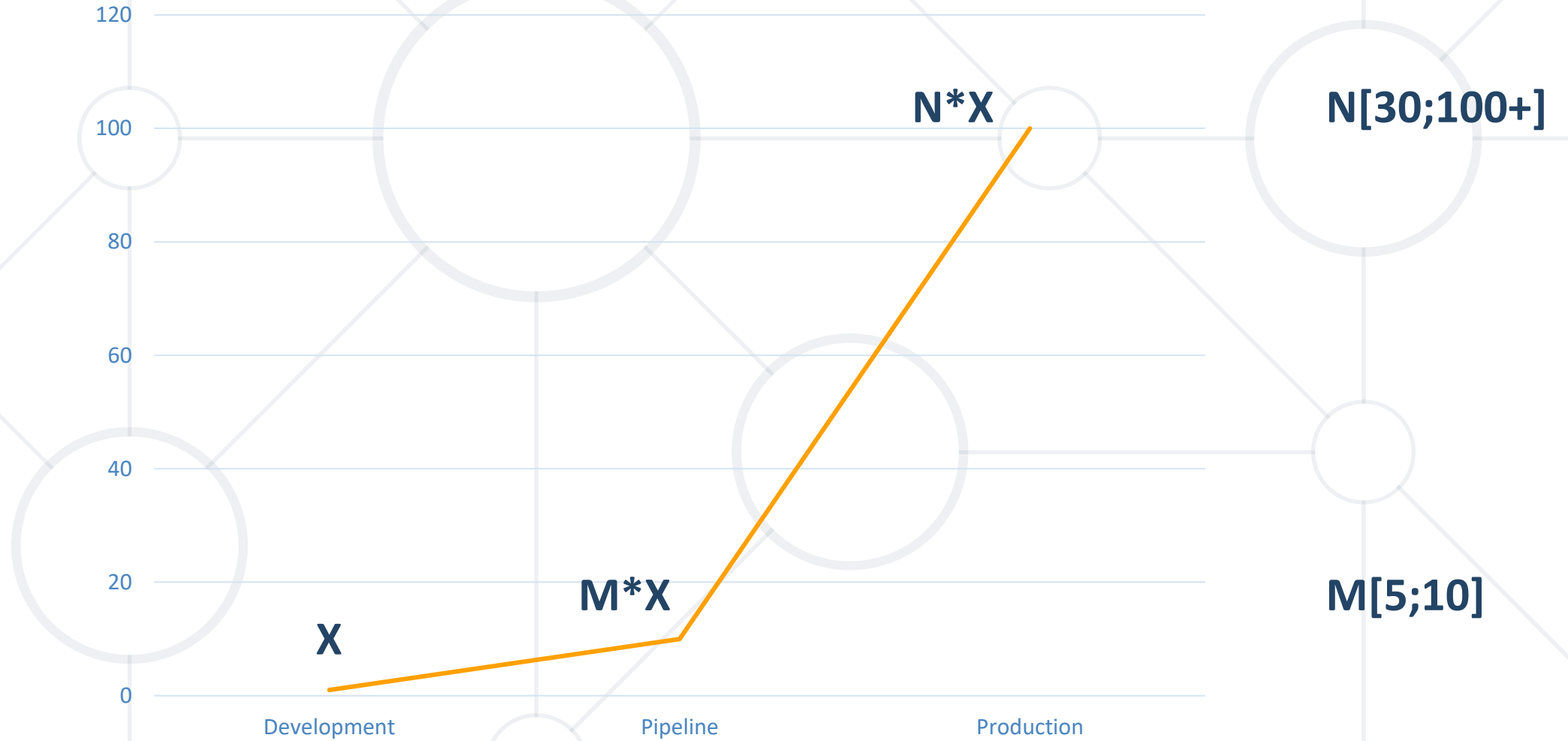


Testing Strategy

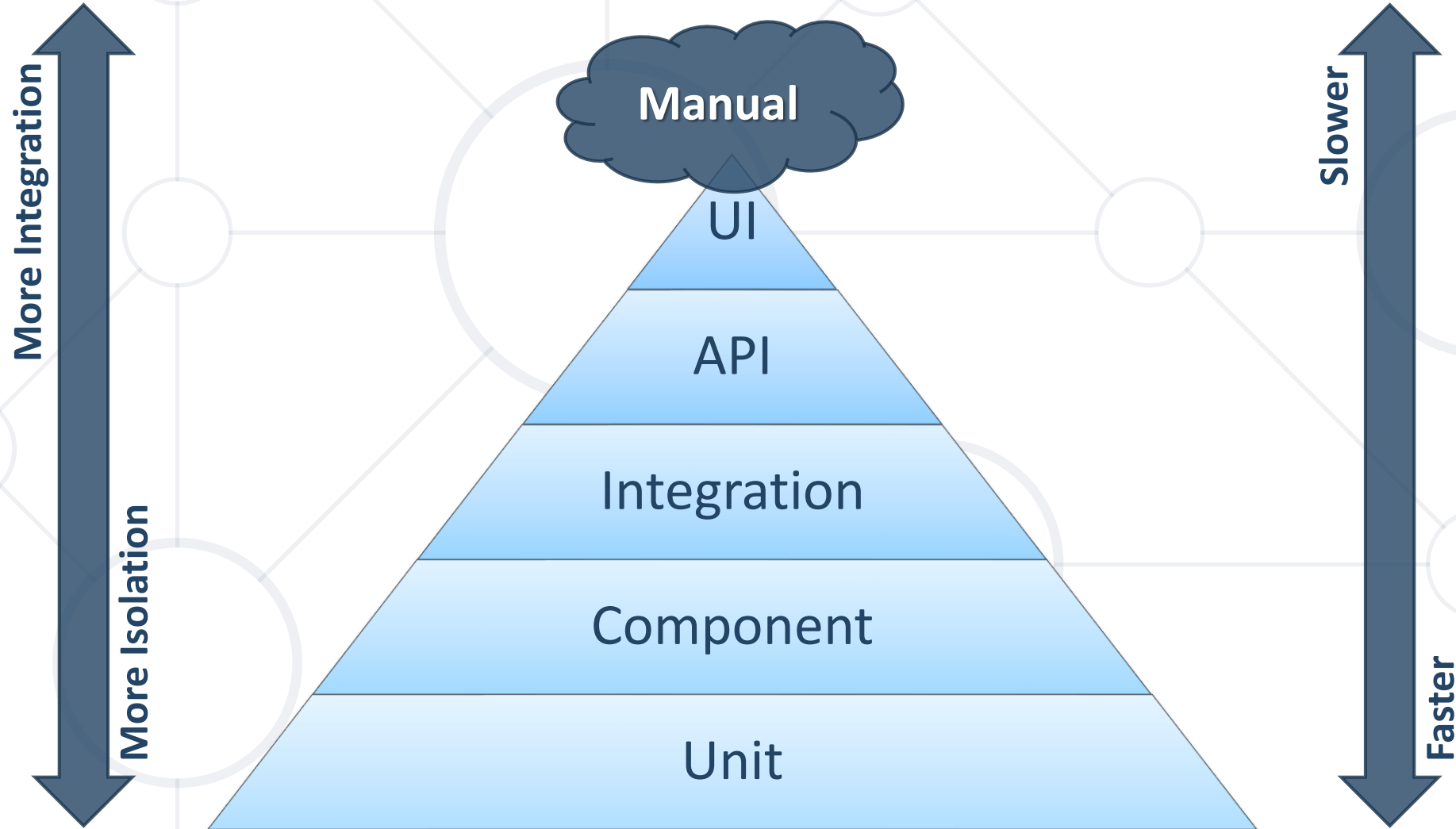
The Why and How

- It *works on my machine* is the enemy of DevOps
- If it isn't tested, it's broken
- Every change must prove it hasn't broken the system before it gets turned into an artefact
- The **Fail-Fast** principle is identifying bugs at the commit stage
- If a test fails, the pipeline **stops immediately**
- If the tests don't pass, the code **doesn't exist** to the rest of the pipeline
- No broken code reaches the registry

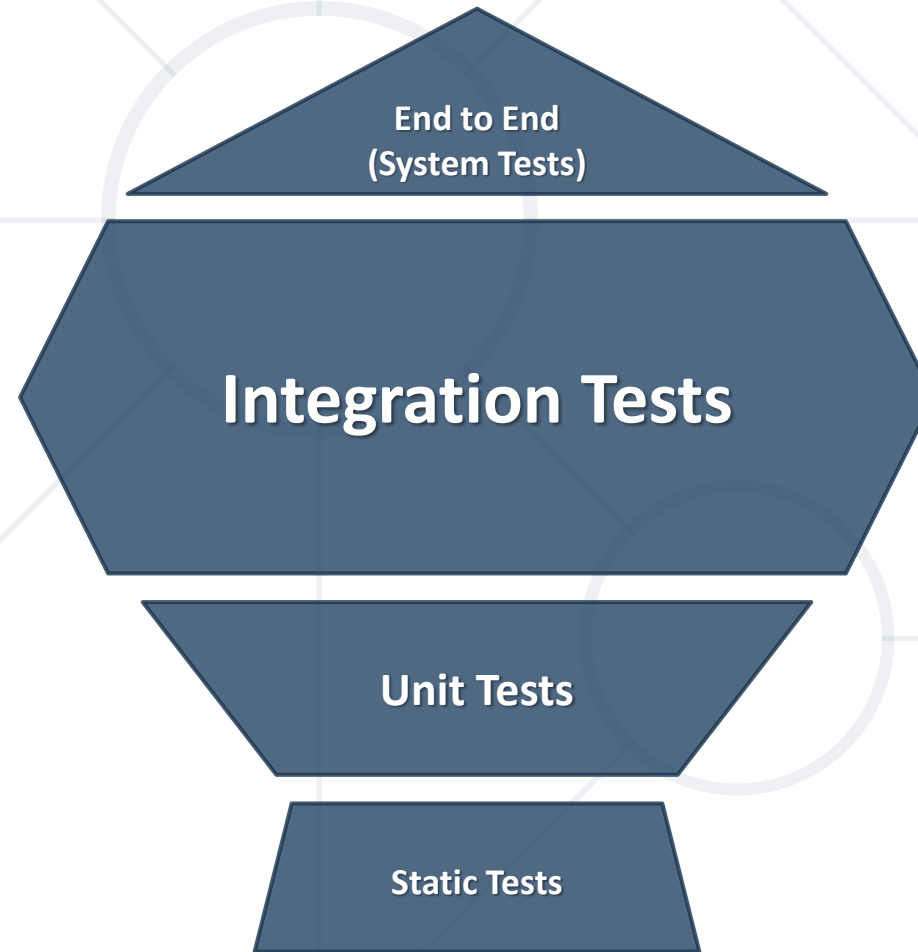
Cost of a Bug



Testing Pyramid



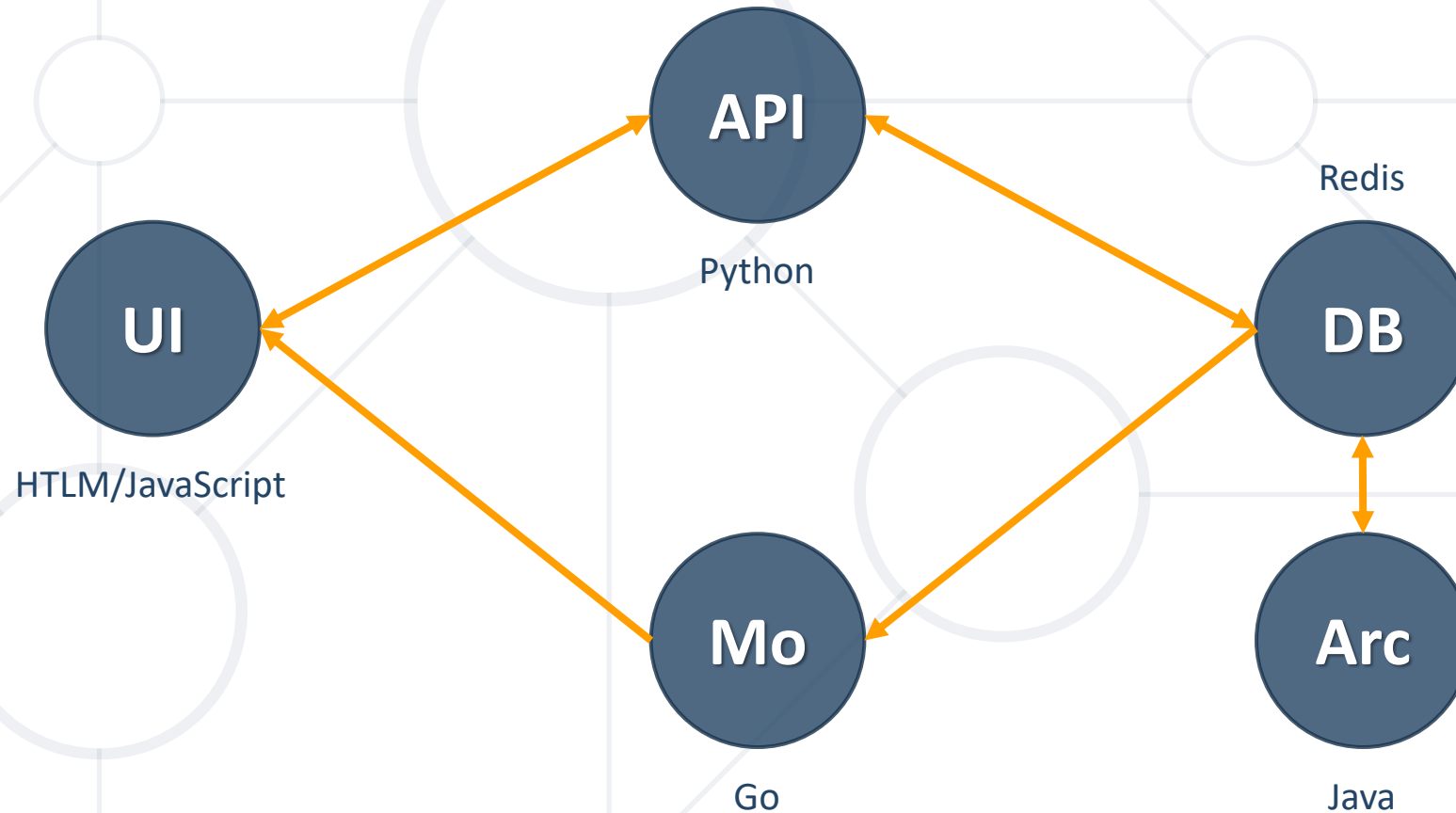
Focus is on **Speed & Cost** through broad base of Unit tests, fewer Integration tests, and minimal UI tests



Focus on **Integration tests** (the "middle")

Ensuring that two services can talk to each other is often more valuable than testing a single function

Our Situation





Practice: Knowledge is the Key
Get Familiar with the Architecture 😊



Test Types

The What

- Testing a single component/function in total isolation
- Cheap and quick to automate and integrate in pipeline
- May involve mocking/abstracting external dependencies
- Popular tools
 - **JUnit, PyTest, unittest, Mocha, Mockito, unittest.mock, etc.**

- Testing the handshake/interaction between components
- For example, can an app connect to a database, and/or insert a record there
- Popular tools
 - **Postman, TestContainers, WireMock, etc.**

- A "shallow" test to ensure the system is up and reachable
- A simplified version of End-to-End testing
- For example, can a browser open the UI, see an element there, and interact with it
- Popular tools
 - **Playwright, Selenium, Cypress, etc.**

- **Static tests** apply code linting; sanity and consistency checks
- **Functional tests** focus on business requirements and verify only the output of an action/functionality
- **End-to-End test** verify that various user flows work as expected
- **Regression tests** implement scenarios that worked in the past and should still be working
- **Acceptance tests** check if a feature/system meets business requirements
- **Performance tests** check for stability and scalability of the product and infrastructure
- **Load testing** is a variation of performance testing that emphasize on behavior under heavy load/stress



Practice: Tests
Let's Create a Few



Automated Testing

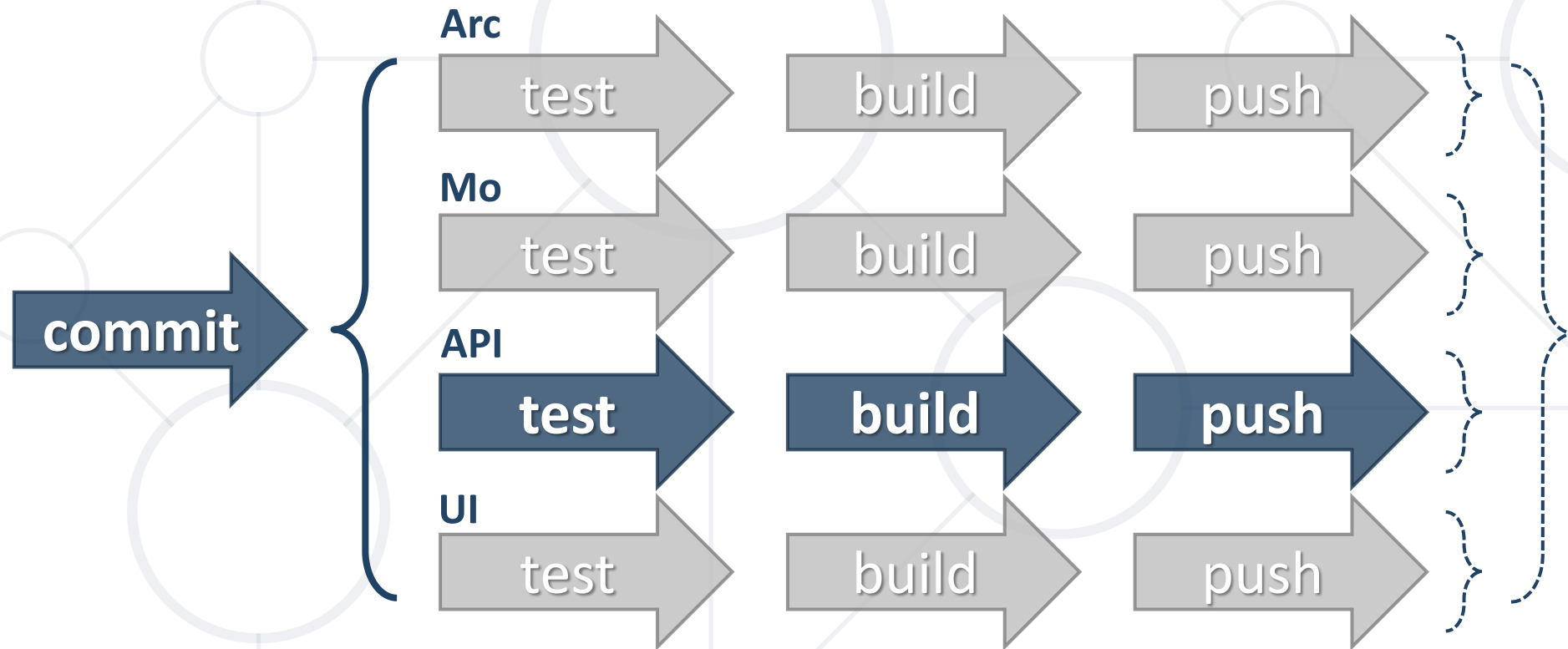
The When and Where

- Testing introduces new dependencies
- Installing those will lead to inevitable conflicts
- We should run each set or set of steps in isolation
- This will
 - Leave our agent/host in clean state
 - Lead to better control over the conditions
- Containerized execution is the solution

- Two ways to organize and govern a multi-service application
 - **Monorepo** – a single repository for all services
 - **Polyrepo** (or multirepo) – each service has its own repository
- We will start with a monorepo
- What is the issue here
 - A change in one service will trigger the building of the rest as well
 - The solution is to use path-based triggering
- This approach will save resources and speed up the feedback

The Green Workflow

Changing one service, will trigger only that portion of the process, being it a separate pipeline, or pipeline portion



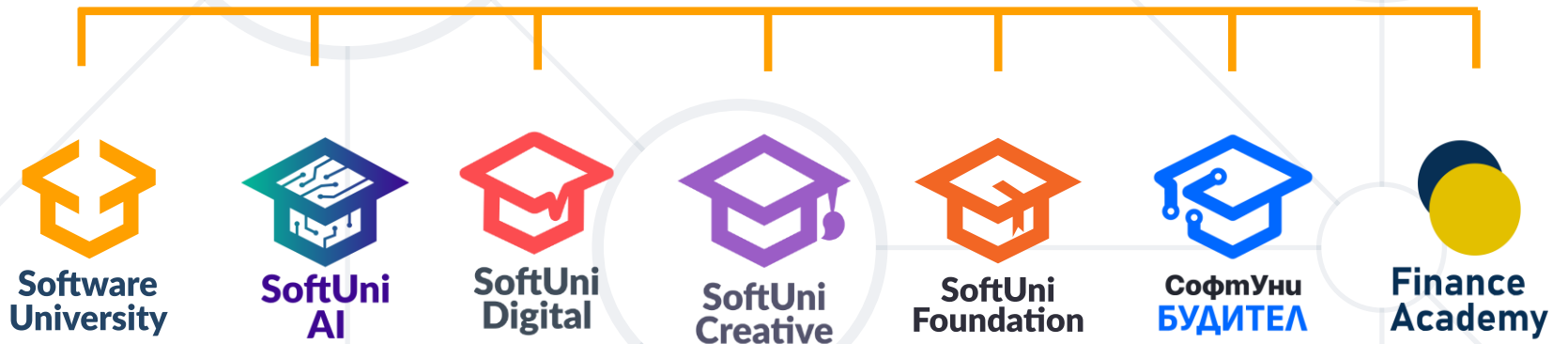


Practice: Automation
Let's Integrate Them in the Pipeline

Questions?



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